

I.2 Reproducing on chat model fine-tuned on narrower domains

To verify that our findings extend beyond the base vs. chat phenomenon, we conducted additional experiments on models fine-tuned in narrower domains. We compare two domain-specific fine-tuning scenarios:

- **Medical domain fine-tuning:** We compare google/gemma-2-2b-it to OpenMeditron/Meditron3-Gemma2-2B from the Meditron3 Sallinen et al. [2025] suite. Crosscoders were trained on 50M tokens from LMSYS and 39M tokens of medical data, including a mixture of [Tran et al., 2024, bio-nlp-umass/bioinstruct], [Chen et al., 2024, FreedomIntelligence/medical-o1-reasoning-SFT], and [Xiong et al., 2024, MedRAG/pubmed].
- **RL fine-tuning on reasoning data:** We compare deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B to nvidia/Nemotron-Research-Reasoning-Qwen-1.5B, which applies extended RL training periods for deeper exploration of reasoning strategies Liu et al. [2025]. Crosscoders were trained on 50M tokens from LMSYS and 50M tokens of reasoning traces from [open-r1/OpenR1-Math-220k](#).

For both comparisons, we trained L1 and BatchTopK crosscoders with comparable $L_0 \approx 100$ on the validation set and measured how many latents are truly specific to the fine-tuned model as determined by Latent Scaling. Table 1 shows results across all investigated models, including the number of fine-tuned-only (FT-only) latents based on the relative norm difference Δ .

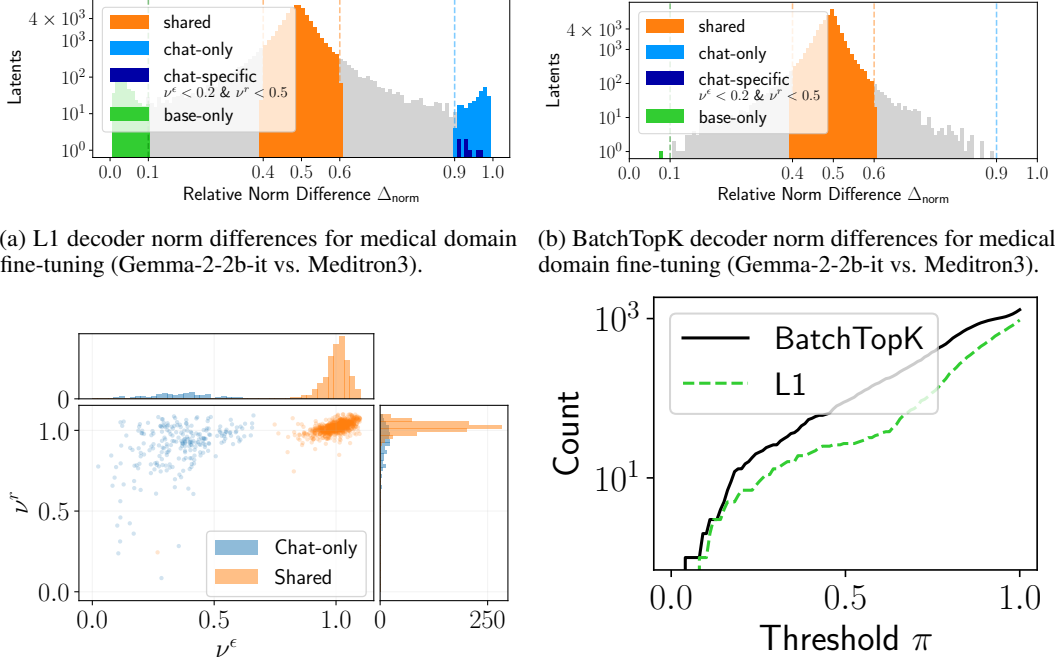
Table 1: Domain-specific fine-tuning results across different model pairs, architectures, and fine-tuning methods. The table shows the systematic pattern where L1 crosscoders consistently misidentify shared latents as fine-tuning-only due to Complete Shrinkage and Latent Decoupling phenomena.

Model	Type	# FT-only ($\Delta \geq 0.9$)	False FT-only ($\nu > 0.6$)	# latents $< \pi$			
				0.2	0.4	0.6	0.8
Gemma2-2B-Chat	BatchTopK	134	1 (0.7%)	301	979	2035	3269
	L1	3176	2132 (67.1%)	13	201	982	2970
Llama-3.1-8B-Chat	BatchTopK	97	13 (13.4%)	382	1263	2073	2848
	L1	2442	1210 (49.5%)	234	765	1594	2440
Llama-3.2-1B-Chat	BatchTopK	17	2 (11.8%)	137	517	1109	1990
	L1	1844	1071 (58.1%)	24	236	790	1330
Qwen-1.5B-Nemotron	BatchTopK	0	0 (0.0%)	0	2	22	127
	L1	59	58 (98.3%)	0	0	2	24
Meditron3-Gemma	BatchTopK	0	0 (0.0%)	13	55	158	529
	L1	246	235 (95.5%)	7	21	35	204

Figure 16 shows the medical domain fine-tuning results, demonstrating the same systematic patterns observed in base vs. chat comparisons. The L1 crosscoder identifies 246 fine-tuning-only latents with $\Delta \geq 0.9$, but 235 of these (95.5%) exhibit high reconstruction ratios $\nu > 0.6$, indicating false attribution due to Complete Shrinkage or Latent Decoupling. In contrast, the BatchTopK crosscoder identifies 0 false fine-tuning-only latents (0.0%).

The reasoning domain comparison (Figure 17) shows even more extreme patterns. For the DeepSeek-R1 vs. Nemotron-Reasoning comparison (Qwen-1.5B-Nemotron), the L1 crosscoder identifies 59 reasoning-related latents as fine-tuning-only with $\Delta \geq 0.9$, but 58 of these (98.3%) exhibit Complete Shrinkage or Latent Decoupling with $\nu > 0.6$ - the highest false attribution rate across all model pairs. The BatchTopK crosscoder again identifies 0 false fine-tuning-only latents (0.0%).

We observe two consistent patterns across all models in Table 1: (i) The Δ metric in L1 crosscoders consistently identifies a large number of latents as fine-tuning-only that actually display Complete Shrinkage or Latent Decoupling, with false attribution rates ranging from 49.5% to 98.3%. (ii) BatchTopK crosscoders maintain low false attribution rates (0.0% to 13.4%) and consistently identify more genuinely fine-tuning-specific latents when using Latent Scaling.



(a) L1 decoder norm differences for medical domain fine-tuning (Gemma-2-2b-it vs. Meditron3).

(b) BatchTopK decoder norm differences for medical domain fine-tuning (Gemma-2-2b-it vs. Meditron3).

(c) L1 error vs reconstruction ratio for medical domain fine-tuning, showing Complete Shrinkage and Latent Decoupling patterns.

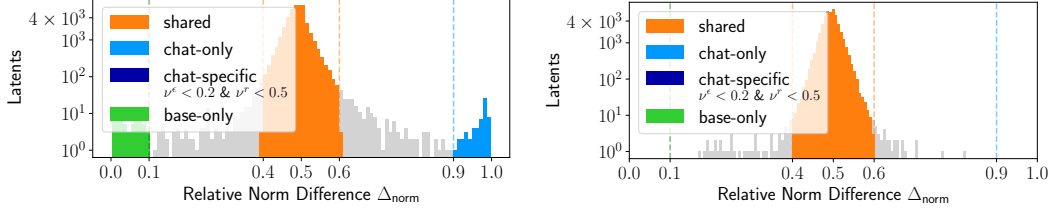
(d) Latents vs threshold comparison for medical domain fine-tuning, comparing L1 and BatchTopK identification of domain-specific latents.

Figure 16: Domain-specific fine-tuning results for medical domain (Gemma-2-2b-it vs. Meditron3-Gemma2-2B). **Top:** Decoder norm differences for L1 (left) and BatchTopK (right) crosscoders. **Bottom:** L1 error vs reconstruction analysis (left) and threshold comparison (right). The results demonstrate that L1 crosscoders systematically misidentify shared medical concepts as fine-tuning-only, while BatchTopK crosscoders more accurately identify genuinely domain-specific latents. Medical fine-tuning was performed on 39M tokens of medical data including bioinstruct, medical reasoning, and PubMed content.

These results demonstrate that our findings reproduce across narrow domain fine-tuning (medical & reasoning), different architectures (Qwen & Llama), and alternative fine-tuning algorithms (RL tuning), supporting the generality and robustness of our analysis.

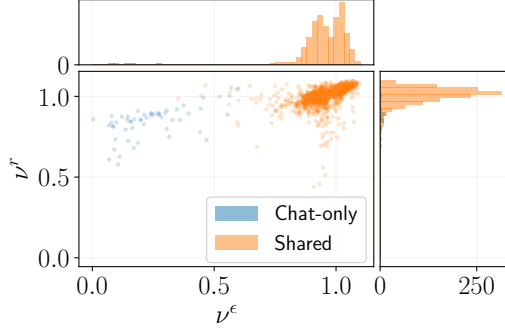
J Reproducing results on independently trained L1 crosscoder

We validate our findings by analyzing a crosscoder independently trained by Kissane et al. [2024a] on the same models and layer than ours. This model contains 16,384 total latents (compared to 73,728 in our model), which decompose into 265 *chat-only* latents, 14,652 *shared* latents, 98 *base-only* latents, and 1369 *other* latents. Figure 18 shows the reconstruction ratio ν^r and error ratio ν^e for all latents, revealing patterns consistent with our previous findings in Figure 2. The overlap between *chat-only* and *shared* latents remains similar - 17.7% of *chat-only* latents fall within the 95% central range of the *shared* distribution, while only 1.1% lie within the 50% central range. We observe even higher ν^e values for *chat-only* latents, suggesting that quite a lot of the *chat-only* latents suffer from Complete Shrinkage. Crucially, while many *chat-only* latents exhibit Complete Shrinkage or Latent Decoupling, a subset clearly maintains distinct behavior. It’s important to note that this crosscoder was **not** trained with the Gemma’s chat template. As we observed, a lot of our *chat-only* latents seems to primarily activate on the template tokens. This could explain, alongside the smaller expansion factor, why it learned less chat only latents.

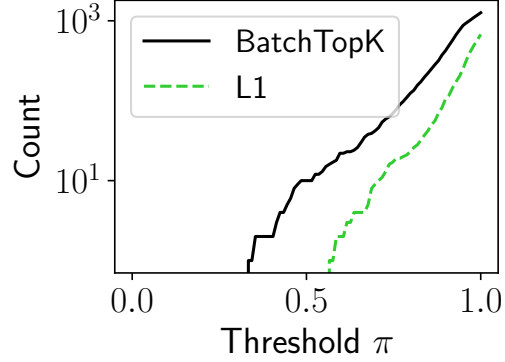


(a) L1 decoder norm differences for reasoning domain fine-tuning (R1dist-Qwen-1.5B vs. Nemotron).

(b) BatchTopK decoder norm differences for reasoning domain fine-tuning (R1dist-Qwen-1.5B vs. Nemotron).



(c) L1 error vs reconstruction ratio for reasoning domain fine-tuning, showing Complete Shrinkage and Latent Decoupling patterns.



(d) Latents vs threshold comparison for reasoning domain fine-tuning, comparing L1 and BatchTopK identification of domain-specific latents.

Figure 17: Domain-specific fine-tuning results for reasoning domain (DeepSeek-R1-Distill-Qwen-1.5B vs. Nemotron-Research-Reasoning-Qwen-1.5B). **Top:** Decoder norm differences for L1 (left) and BatchTopK (right) crosscoders. **Bottom:** L1 error vs reconstruction analysis (left) and threshold comparison (right). The reasoning domain shows the most extreme misattribution patterns, with 98.3% of L1-identified latents being false positives. RL fine-tuning was performed on 50M tokens of reasoning traces from OpenR1-Math-220k.

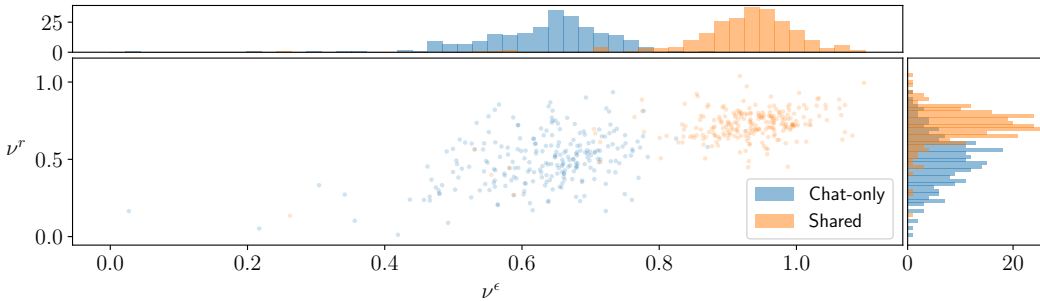


Figure 18: The y -axis is the reconstruction ratio ν^r and the x -axis is the error ratio ν^ϵ . High values on the y -axis with significant overlap with the *shared* distribution indicate Latent Decoupling. High values on the x -axis indicate Complete Shrinkage. We zoom on the ν range between 0 and 1.1.

K Training Details

We trained both crosscoders with the following setup:

- **Base Model:** Gemma 2 2B.
- **Chat Model:** Gemma 2 2B it.

- **Layer used:** 13 (of 26)¹⁴.
- **Expansion factor:** 32, resulting in 73728 latents.
- **Initialization:**
 - Decoder initialized as the transpose of the encoder weights.
 - Encoder and decoder for both models are paired with the same initial weights.
 - The L1 crosscoder is initialized to have a norm of 0.05 while the BatchTopK crosscoder is initialized to have a norm of 1.0. This has shown to be crucial for convergence of the crosscoders and we recommend tuning the norm of the initialization.
 - **Training Data:** 100M tokens from [Fineweb](#) (web data; ODC-By v1.0 License) [Penedo et al., 2023] and [lmsys-chat](#) (chat data; [Custom License](#)) [Zheng et al., 2024], respectively.

As mentioned in Section I.1, for the Llama 3.1 8B BatchTopK crosscoder, we anneal k from 1000 to 200 over 5000 steps. We recommend this to prevent dead latents.

Refer to Table 2 and Table 3 for the training details. We use the tools [nnsight](#) (MIT License) [Fiotto-Kaufman et al., 2024] and [our branch](#) of [dictionary_learning](#) (MIT License) [Marks et al., 2024] to train the crosscoder.

Epoch	μ	LR	Split	FVE (Base)	FVE (Chat)	Dead	Total FVE	L0
1	$4e-2$	$1e-4$	Train	81.5%	82.9%	-	82.3%	112.3
			Val	83.8%	85.2%	7.8%	84.6%	112.5
2	$4.1e-2$	$1e-4$	Train	79.6%	80.7%	-	80.3%	101.7
			Val	83.6%	84.9%	8.1%	84.4%	101.0

Table 2: **L1 crosscoder training statistics.** FVE stands for Fraction of Variance Explained. LR stands for Learning Rate. The L1 regularization parameter μ was slightly increased in the second epoch to improve sparsity, resulting in lower L0 values. We present statistics for both epochs to illustrate this progression.

Epochs	k	LR	Split	FVE (Base)	FVE (Chat)	Dead	Total FVE	L0
2	100	$1e-4$	Train	86.2%	86.9%	-	86.6%	100
			Val	88.1%	87.0%	12.0%	87.6%	99.48

Table 3: **BatchTopK crosscoder training statistics.** FVE stands for Fraction of Variance Explained. LR stands for Learning Rate.

L Additional statistics on the Crosscoders

In this section, we present additional statistics for both the L1 and BatchTopK crosscoders, focusing on the distribution of cosine similarities between decoder latents, latent activation frequencies and the number of [chat-only](#) latents mainly activating on template tokens. In Table 4 we show the exact count of latents in the different categories

Cosine similarity between decoder latents. Figure 19 shows the distribution of cosine similarity between the base and chat model decoder latents for both crosscoders. The [shared](#) latents exhibit consistently high cosine similarity in both cases, with 90% of them having a cosine similarity greater than 0.9 in the L1 crosscoder and 61% in the BatchTopK crosscoder. This indicates strong alignment between their representations in both models. Since the norm of one of the two decoder vectors is ≈ 0 for [base-only](#) and [chat-only](#), these values are less informative.

¹⁴Specifically, we load the model using the `transformers` library from Wolf et al. [2020] and collect the activations from the output of the `model.layers[13]` module

Name	Δ_{norm}	Count	
		L1	BatchTopK
<i>base-only</i>	0.0-0.1	1,437	5
<i>chat-only</i>	0.9-1.0	3,176	134
<i>shared</i>	0.4-0.6	53,569	62373

Table 4: Classification of latents based on relative decoder norm ratio (Δ_{norm}).

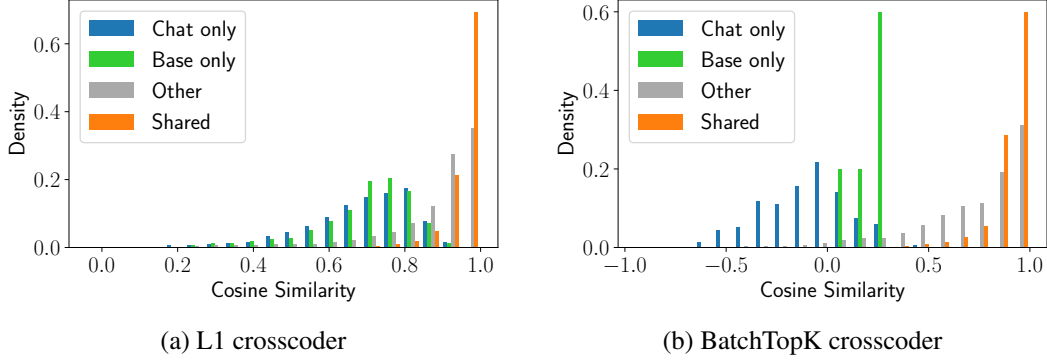


Figure 19: Distribution of cosine similarity between base and chat model decoder latents. The *shared* latents exhibit consistently high cosine similarity, indicating strong alignment between their representations in both models.

Latent activation frequencies. Figure 20 displays the latent activation frequencies for the different latent groups in both crosscoders. Similarly to [Mishra-Sharma et al., 2025], we find that *shared* latents have lower latent activation frequencies than model-specific *base-only* and *chat-only* latents. Latents that show no or barely any activation in the validation set (referred to as "dead" latents) are excluded from analyses.

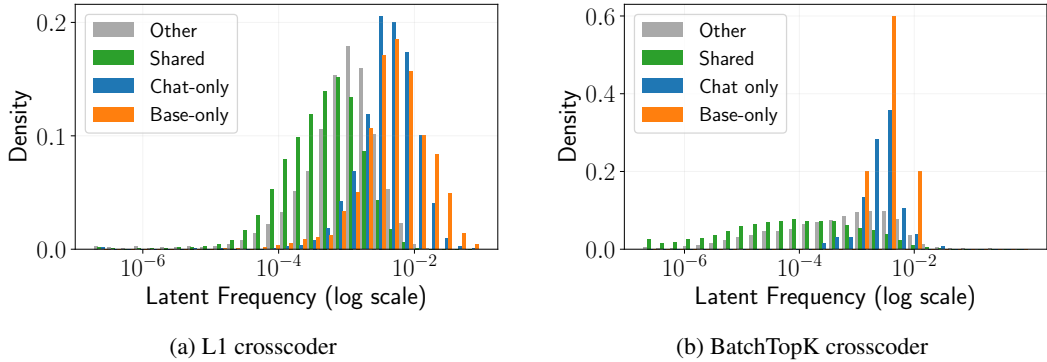


Figure 20: Distribution of latent activation frequency. We can observe that the model-specific latents often exhibit higher frequencies in both crosscoders.

Correlation with ν metrics. We observe a high Spearman correlation between our metrics and latent activation frequency in the L1 crosscoder, especially for ν^r (ν^r : 0.458 and ν^e : 0.83 where $p < 0.05$)¹⁵. We observe no such correlation in the BatchTopK crosscoder. Mishra-Sharma et al. [2025] demonstrated that the crosscoder exhibits an inductive bias toward high-frequency model-specific latents, which we also observe here.

¹⁵Pearson correlation shows less correlation for ν^r (ν^r : -0.02 and ν^e : 0.55) since the relationship is non-linear.